



Solution Set

Question 1

A portfolio has two regimes. Calm regime has mean daily PnL 0.1 million and volatility 1 million. Stress regime has mean -2 million and volatility 5 million. Stress probability is 8 percent. Compute unconditional mean and variance.

Answer key

Mean:

$$\mathbb{E}[\Pi] = 0.92(0.1) + 0.08(-2) = -0.068$$

Use total variance:

$$\begin{aligned}\text{Var}(\Pi) &= \mathbb{E}[\text{Var}(\Pi | R)] + \text{Var}(\mathbb{E}[\Pi | R]) \\ \mathbb{E}[\text{Var}(\Pi | R)] &= 0.92(1^2) + 0.08(5^2) = 2.92\end{aligned}$$

The conditional means are 0.1 and -2, with unconditional mean -0.068, so

$$\begin{aligned}\text{Var}(\mathbb{E}[\Pi | R]) &= 0.92(0.168)^2 + 0.08(-1.932)^2 \approx 0.3246 \\ \text{Var}(\Pi) &\approx 3.2446\end{aligned}$$

Question 2

Prove the tower property

$$\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]$$

Answer key

The conditional expectation $\mathbb{E}[X|Y]$ is the best Y -measurable average of X . Integrating it over all possible Y values restores the unconditional average. In discrete form,

$$\begin{aligned}\mathbb{E}[\mathbb{E}[X | Y]] &= \sum_y \mathbb{E}[X | Y=y] \mathbb{P}(Y=y) \\ &= \sum_y \sum_x x \mathbb{P}(X=x | Y=y) \mathbb{P}(Y=y) = \sum_x x \mathbb{P}(X=x) = \mathbb{E}[X]\end{aligned}$$

Question 3

A loss model has

$$\mathbb{E}[L | S] = 3 + 4S$$

where S is a binary stress indicator with

$$\mathbb{P}(S=1) = 0.1$$

Compute

$$\mathbb{E}[L]$$

Answer key

Use the tower property:

$$\mathbb{E}[L] = \mathbb{E}[\mathbb{E}[L | S]] = \mathbb{E}[3 + 4S] = 3 + 4(0.1) = 3.4$$

This shows how a conditional forecast becomes an unconditional forecast once the state probability is known.

Question 4

Show that if

$$X$$

is independent of

$$Y$$

then

$$\mathbb{E}[X | Y] = \mathbb{E}[X]$$

Answer key

If X is independent of Y , the conditional distribution of X given Y is the same as the unconditional distribution. Therefore its conditional average is unchanged:

$$\mathbb{E}[X | Y=y] = \mathbb{E}[X] \quad \text{for all } y$$

Hence

$$\mathbb{E}[X | Y] = \mathbb{E}[X]$$

Question 5

A desk reports

$$\mathbb{E}[L | L > \text{VaR}_{99}]$$

as expected shortfall. What conditioning event is being used and what caveat applies for discrete losses?

Answer key

The conditioning event is that loss exceeds the 99 percent VaR threshold. For continuous distributions this is straightforward. For discrete distributions, probability mass can sit exactly at VaR, so a rigorous ES definition must handle the boundary mass carefully rather than blindly using $L > \text{VaR}$.

Question 6

Derive the best linear predictor of Y from X in mean-square error form.

Answer key

Minimize

$$\mathbb{E}[(Y - a - bX)^2]$$

First-order conditions are

$$\mathbb{E}[Y - a - bX] = 0, \quad \mathbb{E}[X(Y - a - bX)] = 0$$

Solving gives

$$b = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}, \quad a = \mathbb{E}[Y] - b\mathbb{E}[X]$$

This is the population regression formula.

Question 7

A bond has loss given default LGD and default indicator D . If exposure E is known today, write $\mathbb{E}[\text{loss} | \text{information today}]$.

Answer key

If exposure E and conditional default probability p are known given today's information $\mathcal{F}_t$, then

$$\begin{aligned} \mathbb{E}[\text{Loss} | \mathcal{F}_t] &= E \cdot \text{LGD} \cdot \mathbb{P}(D = 1 | \mathcal{F}_t) \\ &= E \cdot \text{LGD} \cdot p \end{aligned}$$

The important point is that the default probability is conditional on today's information, not necessarily the long-run average.

Question 8

Explain conditional VaR forecasting in a volatility-regime model.

Answer key

If today's regime is known or estimated, tomorrow's loss distribution should condition on that regime. A high-volatility regime produces a wider conditional distribution and larger conditional VaR. The unconditional VaR averages across regimes and may understate risk immediately after volatility jumps.

Question 9

Prove the orthogonality property

$$\mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}])Z] = 0$$

for every

$$\mathcal{G}$$

-measurable

$$Z$$

Answer key

The defining projection property of conditional expectation is

$$\mathbb{E}[XZ] = \mathbb{E}[\mathbb{E}[X | \mathcal{G}]Z]$$

for every $\mathcal{G}$ -measurable Z for which expectations exist. Rearranging gives

$$\mathbb{E}[(X - \mathbb{E}[X | \mathcal{G}])Z] = 0$$

This is why conditional expectation is the best mean-square forecast using information $\mathcal{G}$.

Question 10

A model estimates $P(\text{default} | \text{downgrade}) = 12$ percent and $P(\text{downgrade}) = 5$ percent. What additional information is needed to compute $P(\text{downgrade} | \text{default})$?

Answer key

Bayes' rule gives

$$P(\text{downgrade} | \text{default}) = \frac{P(\text{default} | \text{downgrade})P(\text{downgrade})}{P(\text{default})}$$

We still need the unconditional default probability $P(\text{default})$. Without it, the reverse conditional probability is not identified.

Question 11

Why is $\mathbb{E}[L | \text{stress}]$ not a stress loss scenario by itself?

Answer key

$\mathbb{E}[L | \text{stress}]$ is an average over the stress-state distribution. It is not the worst case and not necessarily a particular realized scenario. A stress scenario should specify concrete shocks; conditional expected loss summarizes the average consequence across many possible shocks inside the stress state.

Question 12

Use total expectation to decompose expected portfolio loss by rating class.

Answer key

If R is rating class,

$$\mathbb{E}[L] = \sum_r \mathbb{E}[L | R = r]P(R = r)$$

This decomposition tells whether expected loss is coming from high conditional severity, large exposure share, or both. It is a standard credit-risk attribution idea.